6.06 Further Dynamics and Kinematics
The techniques of differential equations studied in Pure Core are extended to linear motion under a variable force.

Assumed knowledge
Learners are assumed to know the content of GCSE (9–1) Mathematics and A Level Mathematics. They are also assumed to know the content of Pure Core (Y540 and Y541). All of this content is assumed, but

4. Complex problem solving: Learners could use CAS to perform computation when solving complex problems in mechanics, including those which lead to equations or systems that they cannot solve analytically.

5. Practical mechanics: Learners could use computers and/or mobile phones to enrich practical mechanics tasks, using them for data logging, to create videos of moving objects, or to share and analyse data.

Content of Mechanics (Optional paper Y543)

When this course is being co-taught with AS Level Further Mathematics A (H235) the 'Stage 1' column indicates the common content between the two specifications and the 'Stage 2' column indicates content which is particular to this specification. In each section the OCR reference lists 'Stage 1' statements before 'Stage 2' statements.

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
6.01 Dimensional Analysis			
6.01a	Dimensional analysis	a) Be able to find the dimensions of a quantity in terms of M, L and T, and understand that some quantities are dimensionless. *Includes understanding and using the notation $[d]$ for the dimension of the quantity d.* *Learners are expected to know or be able to derive the dimensions of any quantity for which they know the units. Dimensions of other quantities will be given, or their derivation will be the focus of assessment.*	
6.01b		b) Understand and be able to use the relationship between the units of a quantity and its dimensions.	
6.01c		c) Be able to use dimensional analysis as an error check. *e.g. Verify the relationship that power is proportional to the product of the driving force and the velocity.*	
6.01d		d) Be able to use dimensional analysis to determine unknown indices in a proposed formulation. *e.g. Determine the period of oscillation of a simple pendulum in terms of its length, mass and the acceleration due to gravity, g.*	
6.01e		e) Be able to formulate models and derive equations of motion using a dimensional argument.	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
6.02 Work, Energy and Power			
6.02a 6.02c 6.02b	Work	a) Understand the concept of work done by a force. b) Be able to calculate the work done by a constant force. *The force may not act in the direction of motion of the body and so learners will be expected to resolve forces in two dimensions.*	c) Be able to calculate the work done by a constant f in two dimensions using vectors $(\mathbf{F.x})$ or by a varia force $\left(\int F\mathrm{d}x\right)$ in one dimension only.
6.02d 6.02f 6.02e	Energy	d) Understand the concept of the mechanical energy of a body. *i.e. The kinetic and potential energy.* e) Be able to calculate the gravitational potential energy (mgh) and kinetic energy $\left(\frac{1}{2}mv^2\right)$ of a body.	f) Be able to calculate the kinetic energy of a body u the scalar product $\frac{1}{2}m\mathbf{v.v}$ *Learners may be expected to use the formula* $\mathbf{v.v} = \mathbf{u.u} +$ *in solving a variety of problems, for example in calculating kinetic energy of a body.*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
6.02g 6.02h	Hooke's law		g) Understand and be able to use Hooke's law, in the form $T = \frac{\lambda x}{l}$, for elastic strings and springs. *Includes an informal understanding of when Hooke's law does not apply.* h) Be able to calculate the elastic potential energy $\left(E = \frac{\lambda x^2}{2l}\right)$ stored in a string or spring. *Learners will be expected to state the formula for the elastic potential energy stored in a string or spring unless they are explicitly asked to derive it.*
6.02i 6.02j	Conservation of energy	i) Understand and be able to use the principle of the conservation of mechanical energy and the work-energy principle for dynamic systems, including consideration of energy loss.	j) Extend their knowledge of the principle of the conservation of mechanical energy and the work-energy principle to systems which include elastic strings or springs.
6.02k 6.02l 6.02m	Power	k) Understand and be able to use the definition of power (the rate at which a force does work). *Includes* $\text{average power} = \frac{\text{work done}}{\text{time elapsed}}$. l) Be able to use the relationship between power, the tractive force and velocity ($P = Fv$) to solve problems. *e.g. Motion on an inclined plane.* *Includes maximum velocity and speed.* *Learners will be required to resolve forces in two dimensions.*	m) Be able to calculate the power associated with a variable force in two dimensions using the scalar product $P = \mathbf{F.v}$

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
6.03 Impulse and Momentum			
6.03a 6.03c 6.03b 6.03d	Linear momentum	a) Recall and be able to use the definition of linear momentum in one dimension. b) Understand and be able to apply the principle of conservation of linear momentum in one dimension applied to two particles. *Includes using the formula* $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.	c) Recall and be able to use the definition of in two dimensions including the vector fo d) Understand and be able to apply the prin conservation of linear momentum in two applied to two particles. *Includes using the vector form* $m_1\mathbf{u}_1 + m_2\mathbf{u}_2 = m$
6.03e 6.03f 6.03g 6.03h	Impulse	e) Understand and be able to use the concept of the impulse imparted by a force. f) Be able to use the relationship between the instantaneous impulse of a force and the change in momentum $(I = mv - mu)$. *The instantaneous impulse is the impulse associated with an instantaneous change in velocity.* *Learners will only be required to apply this to instantaneous events in one dimension.* *e.g.* *The direct impact of two smooth spheres.* *An impulsive force acting in the direction of an inelastic string.* *Questions involving collision(s) between particles may include multiple collisions and the conditions under which further collisions occur.*	g) Understand and be able to apply the impu momentum principle in two dimensions i vector form $\mathbf{I} = m\mathbf{v} - m\mathbf{u}$. *e.g.* *The oblique impact of two smooth spheres.* *A smooth sphere with a fixed plane surface.* *An impulsive force acting at an angle to an inela* h) Understand and be able to apply the impu momentum principle for a constant force force $\times$ time or for a variable force in one only as $\int F\,\mathrm{d}t$.

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
6.03i 6.03j 6.03k 6.03l	Restitution	i) Recall and be able to use the definition of the coefficient of restitution, including $0 \le e \le 1$. [*Superelastic collisions are excluded.*] j) Understand and be able to use the terms “perfectly elastic” $(e = 1)$ and “inelastic” $(e = 0)$ for describing collisions. *Learners should know that for perfectly elastic collisions there will be no loss of kinetic energy and for inelastic collisions the bodies coalesce and there is maximum loss of kinetic energy.* k) Recall and be able to use Newton’s experimental law in one dimension for problems of direct impact. e.g. *Between two smooth spheres* $(v_1 - v_2 = -e(u_1 - u_2))$ *and a smooth sphere with a fixed plane surface* $(v = -eu)$*, where* u *and* v *are the velocities before and after impact.*	l) Extend their knowledge to problems involving Newton’s experimental law in two dimensions. *e.g. The oblique impacts of two smooth spheres and a smooth sphere with a fixed plane surface.* *Questions may involve the velocity expressed as a two dimensional vector.*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
6.04 Centre of Mass			
6.04a	Centre of mass		a) Understand and be able to apply the principle that the effect of gravity is equivalent to a single force acting at the body's centre of mass. *Includes understanding that, in terms of linear motion, a rigid body may be modelled by a particle of the same mass at its centre of mass.*
6.04b			b) Be able to find the position of the centre of mass of a uniform rigid body using symmetry, for example a rectangular lamina.
6.04c			c) Be able to determine the centre of mass of a system of particles or the centre of mass of a composite rigid body. *Questions may involve any of the rigid bodies listed in the Formulae Booklet, but will be limited to compound shapes such as a uniform L-shaped lamina or a hemisphere abutting a cylinder with a common axis.* *Includes composition by addition or subtraction, for example a rectangular lamina with a semicircle attached to one side, or a rectangular lamina with a semicircle removed.*
6.04d			d) Be able to use integration to determine the position of the centre of mass of a uniform lamina or a uniform solid of revolution.
6.04e	Rigid bodies		e) Be able to solve problems involving the equilibrium of a single rigid body under the action of coplanar forces. *e.g. Suspension of a rigid body from a given point or problems involving the toppling or sliding of a rigid body placed on an inclined plane*. *May include rigid bodies which are hinged to a surface.* [*Hinged bodies are excluded.*]

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
6.05 Motion in a Circle			
6.05a 6.05b 6.05c	Uniform motion in a circle	a) Understand and be able to use the definitions of angular velocity, velocity, speed and acceleration in relation to a particle moving in a circular path, or a point rotating in a circle, with constant speed. *Includes the use of both ω and $\dot{\theta}$.* b) Be able to use and apply the relationships $v = r\dot{\theta}$ and $a = \frac{v^2}{r} = r\dot{\theta}^2 = v\dot{\theta}$ for motion in a circle with constant speed. c) Be able to solve problems regarding motion in a horizontal circle. *e.g.* *Motion of a conical pendulum.* *Motion on a banked track.* *Problems will be restricted to those involving constant forces but learners will be required to resolve forces in two dimensions.*	
6.05d 6.05e	Motion in a vertical circle	d) Understand the motion of a particle in a circle with variable speed. *In 'Stage 1' Learners will be expected to use energy considerations to calculate the speed of a particle at a given point on a circular path.*	e) Extend their understanding of the motion of a particle in a circle with variable speed to include the radial and tangential components of the acceleration.

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
6.05f			f) Be able to solve problems involving motion round a vertical circle including motion which is not restricted to a circular path. *This is restricted to a combination of motion in a circle and free fall.* *e.g. The subsequent motion of a particle moving on the outside of a smooth circular surface.* *The motion of a particle on a string moving in a vertical circle and then as a projectile.*
6.06 Further Dynamics and Kinematics			
6.06a	Linear motion under a variable force		a) Be able to use $a = \frac{dv}{dt}$ or $a = v\frac{dv}{dx}$ to model the linear motion of a particle under the action of a variable force in one dimension only. *Learners will be required to solve problems in which the corresponding differential equation can be solved by either the method of separation of variables or an integrating factor.*